

ALLinHELP

Company: AllinzucolSmartSystem Lab
Document Type: System Architecture

Version: 1.0
Status: Foundation Architecture

1. Purpose

This document defines the overall technical architecture of **ALLinHELP**, including the major software components, AI services, data flow, integrations, security boundaries, scalability strategy, and deployment approach.

Objective: To build a secure, scalable, cloud-native AI-powered Local Services Operating System capable of supporting millions of users while remaining modular and extensible.

2. Architectural Principles

- ✓ Modular architecture with clearly defined service boundaries.
- ✓ API-first design for web, mobile, and third-party integrations.
- ✓ AI-assisted decision making while preserving user choice.
- ✓ Cloud-native deployment.
- ✓ Security and privacy by design.
- ✓ High availability and fault tolerance.
- ✓ Event-driven communication where appropriate.
- ✓ Observability from day one (logging, metrics, tracing).

3. High-Level System Components

Client Applications

Customer Mobile App: Service discovery, booking, payments, messaging, live tracking, ratings.

Helper Mobile App: Job management, availability, navigation, earnings, profile management.

Web Application: Customer access from desktop browsers & business accounts (*future*).

Admin Dashboard: User management, operations, analytics, configuration, support.

Backend Platform & AI Layer

Backend Microservices: API Gateway, Authentication, User, Helper, Booking, Matching, Pricing, Messaging, Payment, Notification, Review & Reputation, Wallet, Analytics, Fraud Detection, Admin.

AI Platform Domain: Operates as a separate logical domain exposing APIs for Matching, Pricing, Trust, Dispatch, Forecast, Fraud, and Recommendations.

4. External Integrations

Domain	Integrations & Providers
Maps & Location	Google Maps Platform / Mapbox (Geocoding, Reverse Geocoding, Distance Matrix, Route Optimization, Geofencing)
Payments	Paystack, Flutterwave, Stripe <i>(Future International Expansion)</i>
Authentication	Google Sign-In, Apple Sign-In, Facebook Login, Phone OTP, Email Verification
Notifications	Push Notifications, SMS Provider, Email Provider, WhatsApp Business Integration <i>(Future)</i>

5. Logical Architecture



6. Core Service Responsibilities

Service Name	Key Responsibilities & Capabilities
Authentication Service	Registration, Login, OAuth, JWT issuance, password reset, OTP verification, session management.
User Service	Stores customer information (Profiles, addresses, preferences, identity status).

Service Name	Key Responsibilities & Capabilities
Helper Service	Stores helper profiles (Skills, availability, verification, service radius, certifications, performance metrics).
Booking Service	Manages booking lifecycle, scheduling, status transitions, and booking history. Booking Lifecycle States: Requested → Matched → Accepted → En Route → In Progress → Completed (Exception states: Cancelled, Disputed)
Matching Service	Coordinates helper selection. Receives customer request, queries AI Matching Engine, returns ranked helper list.
Pricing Service	Receives booking request, queries AI Pricing Engine, returns recommended pricing range.
Messaging Service	Supports real-time chat, media sharing, read receipts, typing indicators, and voice messaging (future).
Wallet Service	Wallet balance management, transaction logs, withdrawal requests, refunds, commission calculation.
Review Service	Maintains customer/helper ratings, reviews, trust metrics, and performance history.
Payment Service	Payment authorization, escrow handling, settlement, refunds, commission distribution, invoice generation.
Notification Service	Omnichannel dispatch: Push, SMS, Email, and In-app notifications.
Fraud Detection Service	Monitors account anomalies, GPS inconsistencies, payment abuse, review manipulation, duplicate identities.
Analytics Service	Collects marketplace metrics, business KPIs, operational insights, and AI engine performance metrics.

7. AI Services Layer

AI Engine	Inputs	Outputs & Capabilities
AI Adaptive Matching Engine	Customer/Helper location, travel distance, traffic, availability, experience, ratings, skills, trust score.	Ranked helper recommendations.
AI Fair Price Prediction Engine	Service type, duration, distance, demand, urgency, historical pricing.	Minimum, recommended, maximum price range + confidence score.
AI Trust Intelligence Engine	Bookings, reviews, payment records, verification status changes, support cases.	Dynamic, continuously updated trust scores for customers and helpers.
AI Dispatch Engine	Geography, travel times, acceptance probability, load balance.	Optimized notification ordering, minimal travel time, max acceptance rate.
AI Forecast Engine	Historical usage patterns, location data, time indicators.	Predictive demand curves, peak period flags, service shortage warnings, regional trends.

AI Engine	Inputs	Outputs & Capabilities
AI Recommendation Engine	User behavior, service history, promotional campaigns.	Suggested related services, preferred helpers, rebooking prompts, promotions.

8. Data Layer

Storage Technology	Data Domain & Managed Objects
Primary DB (PostgreSQL)	Relational data: Users, Helpers, Bookings, Payments, Reviews, Wallets, Service Categories, Platform Configuration.
Cache (Redis)	In-memory data: User Sessions, Frequently accessed metadata, Temporary booking states, Rate limiting counters.
Object Storage (Cloud S3)	Unstructured media: Profile images, Identity verification documents, Certificates, Job photos, Financial receipts.

9. Security Architecture

Authentication & Encryption

- JWT tokens & OAuth 2.0 framework
- Role-Based Access Control (RBAC)
- TLS protocol for data in transit
- AES encryption for data at rest
- Secure cloud secrets management

Identity Verification & Audit

- Government ID validation
- Live selfie verification
- Face comparison matching (*future*)
- Comprehensive audit logging for administrative actions, financial changes, and security events

10. Scalability & 11. Monitoring / Observability

Scalability Strategy

- Stateless backend service architecture
- Horizontal auto-scaling based on demand
- Database read-replication and connection pooling
- CDN caching for static web assets
- Asynchronous background task processing
- Message queues for event-driven workflows

Observability Metrics

- Centralized application logs & infrastructure metrics
- Real-time API latency & Error rate tracking
- Payment failure monitoring & alert triggers
- AI response times & inference latency
- Booking success & completion rate tracking

12. Disaster Recovery & 13. Future Expansion

Disaster Recovery Strategy

- Automated DB backups with point-in-time recovery
- Multi-zone cloud deployment for fault tolerance

Future Modules Support

- Healthcare, home nursing, & facility management
- Hospitality staffing & property management
- Logistics, delivery, & enterprise workforce

- Infrastructure as Code (IaC) provisioning
- Automated rollback strategy for deployments
- Documented business continuity procedures

- Government service programs & partner APIs
- AI-powered scheduling & Cross-border payments

14. Architecture Success Criteria

- ✓ Capable of serving millions of registered users cleanly.
- ✓ Supporting tens of thousands of concurrent active bookings.
- ✓ Delivering low-latency AI matching and price generation response.
- ✓ Maintaining high platform availability and fault isolation.
- ✓ Enabling independent deployment and iteration of microservices.
- ✓ Allowing new service categories to be introduced with minimal architectural changes.